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India
SMART UTILITY
Week 2026
10 – 14 March 2026, New Delhi

Session : ---------

Transforming Smart Meter Troubleshooting and Solar Integration through Digital Twins and Agentic AI

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ABASTRACT & PROBLEM STATEMENT

Smart meters are the backbone of modern grids, enabling real-time monitoring and billing, yet they face persistent operational hurdles.

Traditional troubleshooting remains reactive, leading to higher operational costs and extended downtime for customers.

Solar integration adds complexity with bidirectional power flows and stringent power quality requirements that legacy systems struggle to manage.

Objective: Shift utilities from reactive maintenance to **proactive intelligent framework** using Digital Twins and Agentic AI for anomaly detection and resolution.

Communication Failures

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Delayed Fault Detection

↓

Operational Inefficiencies

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PROACTIVE AI SOLLUTION

SOLUTION ARCHITECTURE: THREE PILLARS

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Cloud Analytics

- At scale processing of meter & solar telemetry
- Pattern mining for anomaly detection
- Predictive maintenance models & scoring

Agentic AI

- Query Based decision making
- self-healing actions
- Continuous learning & GIS alerts

Block chain Logging

- Tamper-proof event & command audit
- Secure State Verification
- Every action permanently recorded

SYSTEM DATA FLOW

Edge checks (15 -30Min) → Stream Ingestion → Analytics Engine → Agents → Secure Logs

FIVE-LAYER DIAGNOSTIC LOGIC ARCHITECTURE

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LAYER 1: COMM STATE
Weight: 0.20

Connectivity & Authentication

- SIM Active Status
- Network Attach & APN
- Auth Success

LAYER 2: SIGNAL QUALITY
Weight: 0.30

Channel Health Assessment

- RSSI: -140dBm to -44 dBm
- SINR Level
- Latency: <500ms (NAN) / <2000ms (WAN)

LAYER 3: TEMPORAL PATTERNS
Weight: 0.20

Communication frequency Analysis

- Last Time Stamp Check
- Interval: 15-30Min
- Silent Period Threshold: >6hours

LAYER 4: DATA INTEGRITY
Weight: 0.15

Packet & Sequence Validation

- CRC / Checksum
- NTP Sync Accuracy
- Sequence Continuity

Layer 5: POWER QUALITY
Weight: 0.15

Electrical parameter Limits

- Voltage RMS
- Frequency: ± 0.5 – 1 Hz
- PF: 0.95 – 1.0

Weighted Fault Prediction

Score = (L1 x 0.2) + (L2 x 0.3) + (L3 x 0.2) + (L4 x 0.15) + (L5 x 0.15)

High Risk Threshold > Threshold → Proactive Maintenance

SYSTEMATIC ACTION & GIS WORKFLOW

Fault Prediction Trigger

- AI Monitors 5-Layer Diagnostic Meter
- High Risk Flag on Score Threshold Breach
- Predictive Window: 12-48 Hours Before Failure

Field Enablement

- Map based Visualization for Dispatch Teams
- Prioritize intervention bases on risk score
- Route Optimization for Efficient Crew Deployment

GIS-Integrated Alerts

- Instant Dashboard notification
- **Data Points:** Meter ID, GIS Coordinates, Fault Type, Risk score
- Visual Heat map of Affected Areas

Execution Cycle

- Checks Every 15-30 Min (Aligned with Meter Reporting)
- Cloud Model Refresh Daily with Historical Trends
- Incremental Learning from New Signal Data

SOLAR INTEGRATION & NET METERING

Digital Twin for Inverters

Creates a virtual replica of solar inverter & its grid interaction to model behavior and forecast generation in real-time. The twin Continuously syncs voltage and frequency parameters with physical grid.

Key Functions: Models bidirectional power flow and detects anomalies like anti-islanding failures instantly.

Continuous V/f & Power Quality Monitoring

Anti-Islanding Detection

Passive: Monitor voltage /frequency drift and phase imbalance naturally occurring during islanding

Active: Injects small disturbances into the grid connection and analyzes the inverter's response signature.

Alert: Abnormal stability during outage simulation triggers alert.

Outcomes and Benefits

- ***Precise net metering***
- ***Rapid fault isolation***
- ***Safe grid interconnection***
- ***GIS mapped Field action***

BENEFITS AND FUTURE FRAMEWORK

Strategic Benefits

Enhanced Reliability

Early fault prediction reduces downtime, improves SAIDI/SAIFI indices and significantly minimizes reactive truck rolls.

Cost Efficiency and Transparency

Lower O&M cost through predictive workflows. Blockchain logging ensures non-repudiation .

Seamless solar integration

Accurate net metering exports and detection of inverter faults or instability under rapidly fluctuating irradiance.

Future Roadmap

- Automated Case Allocation using AI Insights
- Build digital twin of inverter
- Enhance inverter intelligence
- Layer-5 enhancement

USE CASE / CASE STUDY

Context

~250 HT meter cluster with intermittent comms & data inconsistencies. "**Manual Activity Time 10 am -5pm"**

Early AI Detection(pre failure)

- 5-layer score flagged 38 high risk meters, predicting failure 12-48h in advance.
- Signals: (low RSSI + high latency) . silent>6h. sequence gaps and Continuity failure

GIS Based Action

Heatmap of affected poles, optimized technician routes, meter-wise probable cause.

Analytics Based Response

- Meter-wise analytics
- AI algorithms detects off-peak hours by choosing time blocks with the highest RSSI and lowest latency.

Key Outcomes

- 73%resolved remotely(28/38).
- Field visits reduced from 38 to 10.
- ~62% O&M cost saved, zero customer complaints

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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

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